

Project Video:

https://drive.google.com/file/d/1eX_uzHiDU8CyrwBHyKPbNDc4X0ATADhK/view?usp=drive_link

SQL Code:

```
CREATE DATABASE IF NOT EXISTS railway_management;

USE railway_management;
```

```
DROP TABLE IF EXISTS bookings;
DROP TABLE IF EXISTS schedules;
DROP TABLE IF EXISTS routes;
DROP TABLE IF EXISTS passengers;
DROP TABLE IF EXISTS trains;
DROP TABLE IF EXISTS stations;
DROP TABLE IF EXISTS users;
```

```
-- =====
-- USERS
-- =====
```

```
CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(100) NOT NULL,
    role VARCHAR(20) DEFAULT 'admin'
);
```

```
INSERT INTO users(username, password, role)
VALUES ('admin', '1234', 'admin');
```

```
-- =====
-- STATIONS
-- =====
```

```
CREATE TABLE stations (
    station_id INT AUTO_INCREMENT PRIMARY KEY,
    station_name VARCHAR(100) NOT NULL,
    city VARCHAR(100) NOT NULL
);
```

```
-- =====
-- TRAINS
-- =====
```

```
CREATE TABLE trains (
    train_id INT AUTO_INCREMENT PRIMARY KEY,
    train_no VARCHAR(20) NOT NULL UNIQUE,
    train_name VARCHAR(100) NOT NULL,
    train_type VARCHAR(50) NOT NULL,
    total_seats INT NOT NULL
);
```

```
);
```

```
-- =====  
-- ROUTES (distance column included – required by route.py)  
-- =====
```

```
CREATE TABLE routes (  
    route_id INT AUTO_INCREMENT PRIMARY KEY,  
    source_id INT NOT NULL,  
    destination_id INT NOT NULL,  
    distance FLOAT DEFAULT 0,
```

```
    FOREIGN KEY (source_id)  
        REFERENCES stations(station_id)  
        ON DELETE CASCADE,
```

```
    FOREIGN KEY (destination_id)  
        REFERENCES stations(station_id)  
        ON DELETE CASCADE
```

```
);
```

```
-- =====  
-- SCHEDULES  
-- =====
```

```
CREATE TABLE schedules (  
    schedule_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
    train_id INT NOT NULL,  
    route_id INT NOT NULL,
```

```
    travel_date DATE NOT NULL,  
    departure_time TIME NOT NULL,  
    arrival_time TIME NOT NULL,
```

```
    FOREIGN KEY (train_id)  
        REFERENCES trains(train_id)  
        ON DELETE CASCADE,
```

```
    FOREIGN KEY (route_id)  
        REFERENCES routes(route_id)  
        ON DELETE CASCADE
```

```
);
```

```
-- =====  
-- PASSENGERS  
-- =====
```

```
CREATE TABLE passengers (  
    passenger_id INT AUTO_INCREMENT PRIMARY KEY,  
    passenger_name VARCHAR(100) NOT NULL,  
    phone VARCHAR(20) NOT NULL,  
    email VARCHAR(100)
```

```
);
```

```
-- =====  
-- BOOKINGS  
-- =====
```

```
CREATE TABLE bookings (  
    booking_id INT AUTO_INCREMENT PRIMARY KEY,  
  
    passenger_id INT NOT NULL,  
    schedule_id INT NOT NULL,  
  
    seat_no VARCHAR(20) NOT NULL,  
    booking_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    amount DECIMAL(10,2) NOT NULL,  
  
    status VARCHAR(20) DEFAULT 'Confirmed',  
  
    FOREIGN KEY (passenger_id)  
        REFERENCES passengers(passenger_id)  
        ON DELETE CASCADE,  
  
    FOREIGN KEY (schedule_id)  
        REFERENCES schedules(schedule_id)  
        ON DELETE CASCADE  
);
```

```
-- =====  
-- SAMPLE STATIONS  
-- =====
```

```
INSERT INTO stations(station_name, city)  
VALUES  
( 'Dhaka Railway Station', 'Dhaka'),  
( 'Chittagong Railway Station', 'Chittagong'),  
( 'Sylhet Railway Station', 'Sylhet'),  
( 'Rajshahi Railway Station', 'Rajshahi'),  
( 'Khulna Railway Station', 'Khulna'),  
( 'Mymensingh Railway Station', 'Mymensingh');
```

```
-- =====  
-- SAMPLE TRAINS  
-- =====
```

```
INSERT INTO trains  
(train_no, train_name, train_type, total_seats)  
VALUES  
( '701', 'Subarna Express', 'Intercity', 500),  
( '703', 'Mohanagar Express', 'Intercity', 450),  
( '705', 'Upukol Express', 'Intercity', 400),  
( '707', 'Silk City Express', 'Intercity', 450),  
( '709', 'Tista Express', 'Intercity', 400);
```

```
-- =====  
-- SAMPLE ROUTES (with distance)  
-- =====
```

```
INSERT INTO routes(source_id, destination_id, distance)  
VALUES  
(1,2, 264),  
(1,3, 247),  
(1,4, 256),  
(1,5, 334),  
(1,6, 120);
```

```
-- =====  
-- SAMPLE SCHEDULES  
-- =====
```

```
INSERT INTO schedules  
(train_id, route_id, travel_date, departure_time, arrival_time)  
VALUES  
(1,1, '2026-09-20', '07:00:00', '13:00:00'),  
(2,2, '2026-09-20', '08:00:00', '15:00:00'),  
(4,3, '2026-09-21', '09:00:00', '14:00:00'),  
(3,4, '2026-09-22', '06:30:00', '12:00:00'),  
(5,5, '2026-09-23', '07:30:00', '12:30:00');
```

```
-- =====  
-- SAMPLE PASSENGERS  
-- =====
```

```
INSERT INTO passengers  
(passenger_name, phone, email)  
VALUES  
( 'Rahim Ahmed', '01711111111', 'rahim@gmail.com'),  
( 'Karim Hasan', '01822222222', 'karim@gmail.com'),  
( 'Nusrat Jahan', '01933333333', 'nusrat@gmail.com');
```